

A Comparison of Multilayer Perceptron and Support Vector Machine in Identifying Prediabetic & Diabetic Patients

1. Introduction

Diabetes is a chronic, metabolic disease defined by high concentrations of blood glucose, i.e. blood sugar, which through long periods of time can lead to severe harm to the eyes, kidneys, heart, nerves, and blood vessels [1]. In 2021, over 38.4 million people had diabetes in the United States [2], but thanks to the great amount of progress that's been made in the medical field, diabetic patients are now easily treatable. Nevertheless, it is important to recognise any traits that a diabetic or prediabetic person may exhibit.

In this paper a fair and critical evaluation of two different neural computing methods will be conducted, with the aim of comparing them to determine which model best identifies prediabetic and diabetic patients. The dataset which will allow to do this is taken from [3] and the methods which will be implemented are the Multilayer Perceptron (MLP) and the Support Vector Machine (SVM).

For an initial analysis of the dataset, including basic statistics, please refer to [4].

2. Summary of Methods with Pros & Cons

2.1. Multilayer Perceptron (MLP)

The MLP model can be defined as a supervised (i.e. it uses labelled training data) feed forward neural network, which, as illustrated by Figure 1, is made up of one input layer, at least one hidden layer and an output layer [6]. MLPs are universal approximators, meaning that, if given suitable weights, they can potentially approximate any continuous function with the help of as little as one hidden layer [7-8].

The most efficient way to train a MLP model is with the use of a standard backpropagation (SBP) learning algorithm. After the feed forward propagation run, the SBP algorithm will compute an estimate of the gradient of an error function (generally the mean squared error), with respect to a set of weights and finally backpropagate the error with the scope of subsequently updating the weights accordingly.

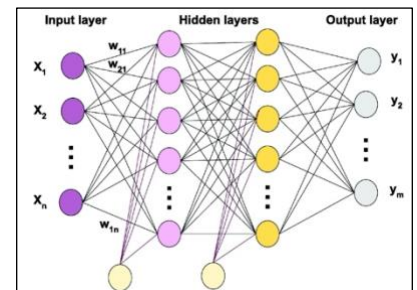


Figure 1. Structure of a MLP model adapted from [5], Fig. 1.

Pros	Cons
Performs well with large amounts of structured input data.	Prone to overfitting, especially when regularisation techniques are not implemented, or the model is too complex.
Automatically learns increasingly complex feature representations by passing input data through multiple layers, hence enabling more effective feature extractions and pattern recognitions [9].	Computationally very time-consuming, in particular for large datasets and deep networks [10].
Highly flexible, meaning MLPs can estimate any continuous function when given at least one hidden layer.	Can suffer from vanishing gradients problem during training, although this can be tackled with the rectified linear unit (ReLU) activation function.

Table 1. Positives and negatives of the MLP model.

2.2. Support Vector Machine (SVM)

The goal of the SVM model is to find a decision boundary which separates the training data

in such a way that the functional margins, dependent on the closest sample(s) in each class (as shown in Figure 2), have maximal width. Since not all datasets are linearly separable in their original feature space, SVMs make use of the kernel trick to project the data into a higher dimension, where a hyperplane can be utilised to separate the classes.

The type of kernel will depend on the type of SVM utilised, for example, if a polynomial learning machine is implemented, then it would be optimal to employ a kernel function of the type $(x^T x_i + 1)^p$, where the power p is previously specified and $i = 1, \dots, N$ [12].

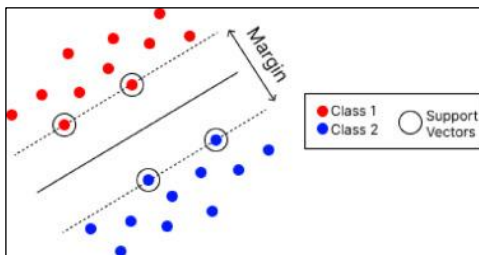


Figure 2. SVM classifier for a linear model adapted from [11], Fig. 7.

The SVM and MLP models differ substantially in their structure – the model complexity of a standard MLP can be managed by limiting the number of features (e.g. the count of hidden neurons), meanwhile the SVM can control model complexity without relying on the dimensionality of the input features [12].

Pros	Cons
Does not tend to overfit, in particular when the right kernel and regularisation are implemented.	Requires careful hyper-parameter tuning, which can be very time-consuming.
Effective in high-dimensional spaces and can handle non-linear data via the kernel trick.	Does not work optimally with large datasets and is very computationally expensive.

Table 2. Positives and negatives of the SVM model [13].

3. Hypothesis Statement(s)

- It is expected that the final version, after hyper-parameter tuning, of both models will perform significantly better than that of the baseline.
- Based on the initial baseline results, it is projected that the SVM's model accuracy will outperform that of the MLP's model, even if by a small margin.
- The training time of the MLP model is anticipated to be notably higher than that of the SVM's model due to the fact that the former has more hyper-parameters to tune than the latter.
- Finally, the two models will output a similar performance for the classification task that's been given to them, i.e. identifying prediabetic and diabetic patients.

4. Description of Choice of Training

This section will guide the reader through the training process of each model and give an understanding of why specific choices (e.g. selection of hyper-parameters) were preferred over others.

Firstly, a 10% subset of the dataset was taken, as it is known that the SVM model can be very computationally expensive and time-consuming to run, in particular with large datasets, so to make a fair comparison of the two, this was done for both models (this subset will be solely used for the baseline model and to run the grid search with the goal of hyper-parameter tuning). That subset was then split between training and testing (80%-20% respectively) and a simple scaling on the data was conducted before running the baseline model. The scaling was done due to the fact that the SVM model greatly relies on dot products and distance measures between data points, therefore results would be skewed if a feature's range was very large. Again, this was done for both models to make a fair comparison of the two methods.

Multilayer Perceptron					
Learning Rate (LR)	Max Epochs	Hidden Neurons	Weight Decay (WD)	Momentum (M)	Optimal Hyper-parameters
[0.001, 0.01, 0.1]	[250, 500]	[15, 30, 45]	[0, 0.001, 0.01]	[0, 0.9]	LR=0.01, Epochs=500, Hidden=30, WD=0.001, M=0

Table 3. Hyper-parameter values implemented in the grid search.

4.1. MLP Structure & Training Method

In the case of the MLP, the baseline model was made up by an input layer, an output layer and a single hidden layer with 21, 1 and 15 neurons respectively. 21 like the number of features, 1 as the target variable can have one of two different outputs and 15 was chosen thanks to one of the rule-of-thumb methods used in [14], that is $\frac{2}{3}(21) + 1 = 15$. The default learning rate was set to 0.001, the optimiser used was the standard gradient descent (SGD), the loss function implemented was the Binary Cross-Entropy Loss (BCELoss) function, which output can be either 0 or 1 as per that of the target variable in the utilised dataset, and a maximum of 50 epochs were run to gauge an understanding of the model's initial performance. Additionally, the ReLU activation function was applied to the output of the hidden layer to avoid the vanishing gradients issue, while a sigmoid function was adopted for the output of the output layer.

Following from the above, to improve the baseline model's performance, a grid search, along with a 3-fold cross-validation (CV) technique, was employed for hyper-parameter tuning. The utilised parameters and their respective values can be explored in Table 3, and, as it can be seen, momentum and weight decay were also added together with the previously mentioned hyper-parameters. It was also experimented to add another hidden layer to the MLP's structure, but the obtained results were not as good as when a single hidden layer was adopted.

Lastly, the optimal hyper-parameters were chosen and utilised in the final MLP model, which was then trained on 80% of the dataset and, subsequently tested on the unseen 20%.

Support Vector Machine				
Kernel Function	C	Gamma	Degree	Optimal Hyper-parameters
'linear'	[0.1, 1, 10, 100]	-	-	Kernel='rbf', C=100, Gamma=0.001, Degree=N/A
'rbf'		[0.001, 0.01, 0.1]	-	
'poly'			[2, 3]	

Table 4. Hyper-parameter values implemented in the grid search.

4.2. SVM Structure & Training Method

In the case of the SVM model, the approach taken was inspired by [15], hence the first step executed after scaling the data was to consider and evaluate the performance of a baseline model with the radial-basis-function (RBF) kernel, that is $K(x, y) = e^{-\gamma \|x - y\|^2}$. Next, to make the fairest possible comparison, a grid search with a 3-fold CV was opted for, as was done for the MLP model. In [15], C-W. Hsu et al. solely use the RBF kernel and span the grid search just over the C and gamma parameters, but to exclude any other possible kernel combination to have a better performance, in this case the grid search also contained different kernel functions (e.g. 'linear' and 'poly'). The optimal hyper-parameters along with all their possible values, can be observed in Table 4.

Finally, the best hyper-parameter values were applied to the SVM model, which was trained on 80% of the data and thereafter, tested on the remaining 20%.

Method	MLP			SVM		
Model Type	Baseline	Grid Search	Final Model	Baseline	Grid Search	Final Model
Accuracy	65.63	73.20	75.20	73.13	73.76	75.25
Precision	66.87	72.22	73.00	71.37	71.71	71.90
Recall	61.95	75.39	80.00	77.23	78.50	82.88
F1 Score	64.32	73.77	76.34	74.18	74.95	77.00
Training Time (s)	2.69	-	144.49	0.59	-	69.31

Table 5. Evaluation metrics for both models.

5. Evaluation Methodology

The problem tackled in this report is of the classification type, that is, the model which best classifies prediabetic and diabetic patients. Therefore, the below-mentioned evaluation techniques and measures are based off a key paper which deals with general classification metrics utilised in the field [16]. The evaluation metrics used after running each model, i.e. the baseline, the best grid search model and the final model (where the whole dataset was used) are accuracy, precision, recall, f1 score, and training time. Additionally, a confusion matrix, which displays the count of correctly and wrongly classified true/false positives/negatives and the area under the receiver-operating-characteristic (ROC) curve, which shows the model's capacity to correctly rank predictions (the closer to 1.0, the better the model) were also constructed, to gauge a visual understanding of each method's performance. All the percentages for the metrics are displayed in Table 5, while Figures 3 and 4 show the confusion matrices and the area under the two ROC curves respectively.

6. Analysis & Critical Evaluation of Results

The purpose of the following section will be to analyse the results in order to give them a proper meaning so that the reader can clearly understand what they are being shown. This will be done by providing an analysis and critical evaluation of both models' outcomes and considering the merits and limitations of the analysis.

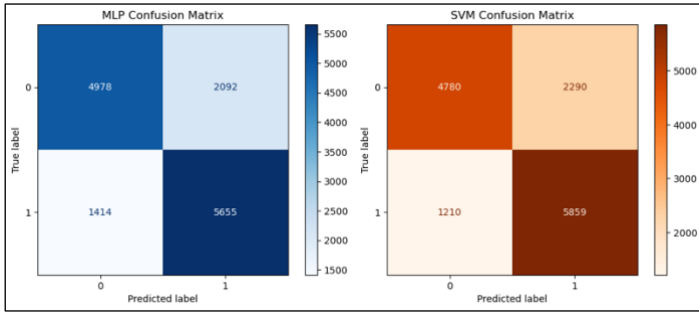


Figure 3. The final models' confusion matrices for the MLP and SVM methods.

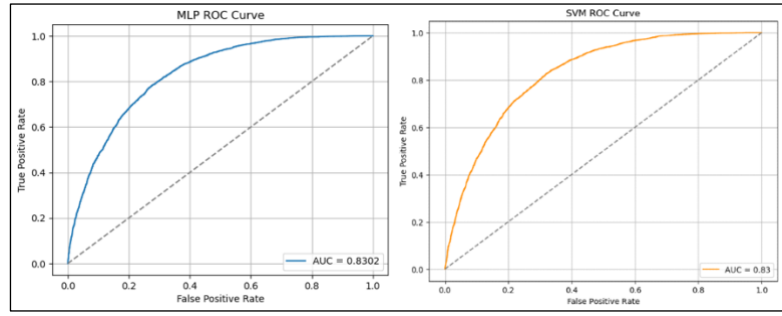


Figure 4. The final models' area under the ROC curves for the MLP and SVM methods.

6.1. Performance of MLP vs SVM

Table 5 displays the performance of each model by using various evaluation metrics. Before analysing the models' efficiencies, it is worth reminding the reader that the baseline and best grid search models were trained using only a 10% subset, meaning they were actually trained on 8% and tested on the remaining 2%. Moreover, another noteworthy remark to make is the fact that the training time for both grid search models was omitted as it would not have been a fair comparison because the number of parameters spanning the MLP grid search model was different to that spanning the SVM's. The grid search in the MLP model summed to a total of 324 fits, while that of the SVM model added to just 120 fits. Therefore, it is predictable for the former to take a longer amount of time than the latter.

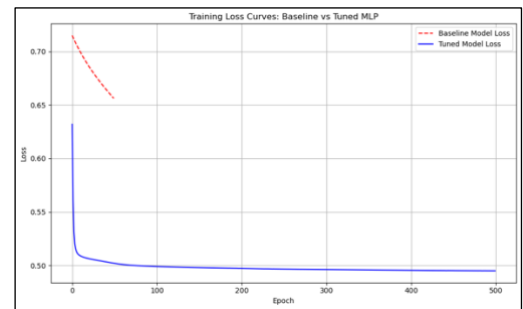


Figure 5. MLP's training loss curves for baseline vs final model.

Let's now observe some of the accuracy values displayed in Table 5. Comparing MLP's baseline with its final model, it can be seen that, as expected, the final model has a higher accuracy than its baseline, by about 10%. This is still the case for the SVM's baseline and final models, even if the margin of improvement between the two is of roughly just 2%. This is because the baseline model for the SVM already performs fairly well, while MLP's baseline accuracy measure is quite low, hence it has a greater margin of improvement. The reason for this may be the fact that in the MLP's baseline model only 50 epochs were implemented, which meant that the model stopped early, as is shown by the red training loss curve in Figure 5. Additionally, another explanation is the fact that the baseline model was only trained on 8% of the dataset, while the final model was trained on 80%.

Both grid search models, when compared with their respective final model, demonstrate very similar behaviour, meaning that the accuracy difference between the grid search and final models is of roughly 2% for both the MLP and SVM methods. This is certainly due to the usage of the 10% subset for the training and testing of the best grid search model, differently from the whole dataset being used for training and testing of the final model.

The MLP's final model accuracy is just 0.05% lower than that of the SVM's final model, which in these cases is insignificant, but the fact that the training time for the SVM model takes significantly less than that of the MLP's, is the first clear sign that, in this case, the SVM method is preferred over the MLP.

The two confusion matrices, which can be observed in Figure 3, make it easy to visualise the difference in the correctly and wrongly classified labels in each model. In both the MLP and SVM confusion matrices, the count of correctly classified patients is greater than that of the wrongly classified ones, as expected. Moreover, the number of correctly classified labels in the SVM model is higher than that of the MLP's by just 6 labels, and vice-versa the number of wrongly classified patients in the MLP model is higher than that of the SVM's by a count of

just 6 labels. The most relevant difference between the two models is the fact that the number of false negatives in the MLP model is considerably higher than that of the SVM's. This is of particular significance in the medical context because if a patient is not correctly diagnosed, meaning they are told they are not suffering from diabetes or prediabetes when they actually are, it could lead to serious complications when untreated (e.g. heart attacks and strokes [17]). Therefore, in this case the SVM model would be preferred over the MLP model as it does a better job in correctly classifying patients, while minimising the error. The area under the receiver-operating-characteristic curves (AUC-ROC) shown in Figure 4 for the MLP and SVM models is 0.8302 and 0.83 respectively, meaning that both models are fairly adequate at ranking predictions, which can also be observed from the confusion matrices in Figure 3. The difference in the AUC-ROC values is not significant enough to confidently say that the SVM model outperforms the MLP model in classifying prediabetic and diabetic patients, or vice versa. Therefore, both models are similarly effective in their classification performance on this dataset, and neither is considerably better than the other, as expected.

6.2. Merits & Limitations of Analysis

The depth with which the evaluation was executed is one of the biggest merits of the above analysis. The most neutral possible comparison was carried out between the two models by using multiple assessment metrics, the most relevant ones being accuracy and f1-score. In order to ensure an unbiased comparison, all data splits were replicated for both models and a grid search with the same number of CV folds (3) was run in each model. Furthermore, due to the fact that the analysis was made on a realistic dataset which had people being surveyed, this analysis could also have some important real-world applications in the classification of diabetic and prediabetic patients, even if ideally the percentages in the evaluation metrics should be higher.

The biggest limitation in the analysis concerns the implementation of the grid search; due to timing and computational constraints, only a very limited number of values were chosen per parameter, for example in the SVM model four values were selected for the C parameter and only three for gamma, meanwhile in the MLP model just two values were picked for the momentum parameter and 3 different learning rates were chosen. This is to say that it may well be that this grid search missed the actual optimal hyper-parameter values, due to the fact that the tuning was not exhaustive.

7. Lessons Learned & Future Work

The most significant lesson learned that can be taken from this report is the fact that there will not necessarily always be a better performing model, it may happen, as it has in this case, that two models perform quite similarly, even if they distinguish themselves under various aspects in their structure and training process. Additionally, it was also learned that in order to make the fairest possible comparison between different models the number of epochs must be the same in both. This was not done for MLP's baseline and final models (50 and 500 epochs respectively), hence the outcomes of the evaluation metrics could have been different.

Following this report, the future work that can be done is to explore further the MLP method by applying different types of loss functions, not only the BCELoss function. Moreover, also trying to differentiate in the optimiser (e.g. Adam) and activation functions (e.g. SoftMax) from layer to layer, could give different results. While, for the SVM model, diverse kernel type functions could be investigated, one of these being the sigmoid kernel.

8. References

- [1] World Health Organization, "Diabetes," *World Health Organization*, 2024. https://www.who.int/health-topics/diabetes#tab=tab_1.
- [2] National Institute of Diabetes and Digestive and Kidney Diseases, "Diabetes Statistics," *National Institute of Diabetes and Digestive and Kidney Diseases*, 2024. <https://www.niddk.nih.gov/health-information/health-statistics/diabetes-statistics>.
- [3] "Diabetes Health Indicators Dataset," *www.kaggle.com*. https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset/data?select=diabetes_binary_5050split_health_indicators_BRFSS2015.csv.
- [4] N. Finzi & S. Elangovan, "Comparison between MATLAB and Python MLP implementation", unpublished manuscript, INM427 Neural Computing, City St. George's University of London, London, UK, 2025.
- [5] K. Y. Chan *et al.*, "Deep neural networks in the cloud: Review, applications, challenges and research directions," *Neurocomputing*, vol. 545, p. 126327, Aug. 2023, doi: <https://doi.org/10.1016/j.neucom.2023.126327>.
- [6] T. Kotsiopoulos, P. Sarigiannidis, D. Ioannidis, and D. Tzovaras, "Machine Learning and Deep Learning in smart manufacturing: The Smart Grid paradigm," *Computer Science Review*, vol. 40, p. 100341, May 2021, doi: <https://doi.org/10.1016/j.cosrev.2020.100341>.
- [7] A. d'Avila Garcez & K. Ryan, 2025. Lecture 2 – The Multilayer Perceptron & Backpropagation Algorithm. Lecture notes distributed in INM427 Neural Computing. 6 February, City St. George's, University of London.
- [8] K. Hornik, "Approximation capabilities of multilayer feedforward networks," *Neural Networks*, vol. 4, no. 2, pp. 251–257, 1991, doi: [https://doi.org/10.1016/0893-6080\(91\)90009-t](https://doi.org/10.1016/0893-6080(91)90009-t).
- [9] S. Banerjee, "Exploring the Power and Limitations of Multi-Layer Perceptron (MLP) in Machine Learning," *Medium*, Apr. 06, 2024. <https://shekhar-banerjee96.medium.com/exploring-the-power-and-limitations-of-multi-layer-perceptron-mlp-in-machine-learning-d97a3f84f9f4>.
- [10] "What is Multilayer Perceptron," *Deepchecks*, Jun. 13, 2024. <https://www.deepchecks.com/glossary/multilayer-perceptron/>.
- [11] H. Ahmetoglu and R. Das, "A comprehensive review on detection of cyber-attacks: Data sets, methods, challenges, and future research directions," *Internet of Things*, vol. 20, p. 100615, Sep. 2022, doi: <https://doi.org/10.1016/j.iot.2022.100615>.
- [12] S. Haykin *et al.*, "Neural Networks and Learning Machines Third Edition." Available: <https://dai.fmph.uniba.sk/courses/NN/haykin.neural-networks.3ed.2009.pdf>.
- [13] K. Dhiraj, "Top 4 advantages and disadvantages of Support Vector Machine or SVM," *Medium*, Sep. 25, 2020. <https://dhirajkumarblog.medium.com/top-4-advantages-and-disadvantages-of-support-vector-machine-or-svm-a3c06a2b107>.
- [14] J. Heaton, "The Number of Hidden Layers," *Heaton Research*, Dec. 28, 2018. <https://www.heatonresearch.com/2017/06/01/hidden-layers.html>.
- [15] C-W. Hsu, C-C. Chang, and C-J. Lin, "A Practical Guide to Support Vector Classification," 2003. Available: <https://www.csie.ntu.edu.tw/~cjlin/papers/guide/guide.pdf>.
- [16] Ž. Đ. Vujovic, "Classification Model Evaluation Metrics," *International Journal of Advanced Computer Science and Applications*, vol. 12, no. 6, 2021, doi: <https://doi.org/10.14569/ijacsa.2021.0120670>.
- [17] Diabetes UK, "Complications of diabetes," *Diabetes UK*, 2024. <https://www.diabetes.org.uk/about-diabetes/complications>.